

Induction: Gauss S 10

$$\begin{array}{l} \textcircled{1} + 2 + \dots + 100 = S \\ \textcircled{100} + 99 + \dots + 1 = S \end{array}$$

$$101 + 101 \dots + 101 = \underline{2S} = 101 \cdot 100$$

~~~~~  
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$$S = \frac{101 \cdot 100}{2}$$

WTP:  $\forall n \in \mathbb{Z}_+$

$$1 + 2 + \dots + n = \sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$n=1$

$$1 = \frac{1(1+1)}{2} = 1 \quad \text{P}(n)$$

$n=2$

$$1+2 = \frac{1+2}{2} + 2 = 2\left(\frac{1}{2}+1\right) = 2\left(\frac{3}{2}\right)$$

$n=3$

$$1+2+3 = \frac{2(3)}{2} + 3 = 3\left(\frac{2}{2}+1\right) = 3\left(\frac{4}{2}\right)$$

$$1+2+3+4 = \frac{3 \cdot 4}{2} + 4 = 4\left(\frac{3}{2}+1\right) = 4\left(\frac{5}{2}\right)$$

Proof. Proceed by induction. The statement is  
already true for  $n=1$  b/c  $1 = \frac{1(2)}{2} = 1$ .

<sup>"Assume  $P(n)$ "</sup>  
Assume that we already know the statement  
for  $n$ . IE assume that  $\leftarrow$  this is  $P(n)$ .

$(1 + 2 + \dots + n = \frac{n(n+1)}{2})$

Adding  $n+1$  to both sides gives

$$1 + 2 + \dots + n + (n+1) = \frac{n(n+1)}{2} + (n+1).$$

The LHS of this is the LHS of  
what we want to prove. The RHS is

$$(n+1) \left( \frac{1}{2} + 1 \right) = (n+1) \left( \frac{n+2}{2} \right) = (n+1) \frac{(n+1+1)}{2}!$$

We conclude that  $\forall n \in \mathbb{Z}_{\geq 0}$ ,

$$1 + 2 + \dots + n \geq \frac{n(n+1)}{2}. \quad \square$$

~~to~~ is  
a proof that  
 $P(n) \Rightarrow P(n+1)$ .

Let  $P(n)$  be a statement which depends on some integer (usually pos)  $n$ .

(Ex.  $P(n) = "1 + 2 + \dots + n = \frac{n(n+1)}{2}"$ )

Goal: Prove  $P(n) \forall n \in \mathbb{Z}_{>0}$ .

Step 1: Prove  $P(1)$  "Base Case"

Step 2: Prove " $P(n) \Rightarrow P(n+1)$ " "Inductive Step"

"Induction" =  $P(1) \wedge \underbrace{P(n) \Rightarrow P(n+1)}_{\forall n \in \mathbb{Z}_{\geq 0}, P(n)} \Rightarrow$

$P(1), P(2), P(3), P(4), \dots, P(n), P(n+1), \dots$

Waitzgo:  $P(n) \neq \boxed{\frac{n(n+1)}{2}}$  ← not "t or F"

$$P(n) = 1+2+\dots+n = \frac{n(n+1)}{2}$$

Note:  $n$  is just a variable.

$$P(1) \wedge P(n) \Rightarrow P(n+1)$$

$$P(a) \Rightarrow P(a+1)$$

$$P(a-1) \Rightarrow P(a)$$

"prove the following case!"



Voraussetzung:  $P(0) \wedge P(n) \Rightarrow P(n+1)$

$$P(2) \wedge P(n) \Rightarrow P(n+1)$$

$$P(2) \wedge P(n) \Rightarrow P(n+2)$$

$$\Rightarrow \forall n \in \mathbb{N}_{\geq 0}, P(n)$$

$$P(-1) \wedge P(n) \Rightarrow P(n+1) \quad |$$

$$\forall n \in \mathbb{Z}_{\geq 0}, P(n)$$

$P(-1), P(-2), P(-3)$



Define: a sequence  $a_1, a_2, \dots, a_n, \dots$

"recursive defn"

$$a_1 = 2 = 2^1$$

$$a_n = 2 \cdot a_{n-1}$$

$$a_n = 2 \cdot a_{n-1}$$

$$a_{n+1} = 2 \cdot a_n$$

$$a_2 = 2 \cdot a_1 = 2 \cdot 2 = 4 = 2^2$$

$$a_3 = 2 \cdot a_2 = 2 \cdot 4 = 8 = 2^3$$

$$a_4 = 2 \cdot a_3 = 2 \cdot 8 = 16 = 2^4$$

Claim:  $\forall n \in \mathbb{Z}_{\geq 0}$ ,  $a_n = 2^n$ .

$$P(n) = "a_n = 2^n"$$

Proof: Proceed by induction.

Base case:  $P(1) \text{ is } "a_1 = 2", i.e., "2 = 2."$

( $P(n) \Rightarrow P(n+1)$ ) This is true.

Inductive step: Assume  $P(n)$ , i.e.,  $a_n = 2^n$ ,

(wtp  $P(n+1)$ , i.e.,  $a_{n+1} = 2^{n+1}$ ) Then  $a_{n+1} = 2 \cdot a_n$

by the defn. of  $a_n$ . Then  $a_{n+1} = 2 \cdot 2^n = 2^{n+1}$ .

Thus  $P(n+1)$  is true.  $\square$

Define:  $a_1 = 0$

$$a_n = \sqrt{3 + 2a_{n-1}}$$

Prove:  $\forall n \in \mathbb{Z}_{>0},$

$$a_n < 3$$

$$P(n) = "a_n < 3"$$

$$a_1 = 0$$

$$a_2 = \sqrt{3 + 2 \cdot 0} = \sqrt{3}$$

$$a_3 = \sqrt{3 + 2a_2} = \sqrt{3 + 2\sqrt{3}}$$

$$a_4 = \sqrt{3 + 2a_3} = \sqrt{3 + 2\sqrt{3 + 2\sqrt{3}}}$$

Proof: Proceed by induction.

Base case:  $P(1) \equiv "a_1 < 3", \text{ i.e., } "0 < 3"$ .

This is true.

I.S. Assume  $P(n)$ , i.e., " $a_n < 3$ ".

(WTP:  $P(n+1)$ , i.e., " $a_{n+1} < 3$ ") Then, by defn,

$a_{n+1} = \sqrt{3 + 2a_n}$ . Since  $a_n < 3$ ,

$$a_{n+1} = \sqrt{3 + 2a_n} < \sqrt{3 + 2 \cdot 3} = \sqrt{3+6} \\ = \sqrt{9} = 3$$

Thus  $a_{n+1} < 3$ .  $\square$

Claim:  $1 + 2 + 4 + 8 + \dots + 2^{n-1} = 2^n - 1$

Proof: Proceed by induction.  $\forall n \in \mathbb{Z}_{\geq 0}$

BC:  $P(1)$  is " $1 = 2 - 1$ ". This is true.

IS: Assume  $P(i)$ , i.e.,  $1 + 2 + \dots + 2^{i-1} = 2^i - 1$ .

(wtp  $P(i+1)$ , i.e.,  $1 + 2 + \dots + 2^{i+1-1} = 2^{i+1} - 1$ )

$$1 + 2 + \dots + 2^{i-1} + 2^i = 2^i - 1 + 2^i = 2^{i+1} - 1$$



Adding  $a^i$  to each side gives

$$1 + a + \dots + a^{i-1} + a^i = a^i - 1 + a^i.$$

The LHS is the LHS of  $P(i+1)$ .

The RHS is  $a - a^i - 1 = a^{i+1} - 1$ .

This is the RHS of  $P(i+1)$ .  $\square$



Claim:  $\forall n \in \mathbb{Z}_{\geq 0}, \underbrace{3 \mid 4^n - 1}_{P(n)}.$

$P(n)$

Proof: Proceed by induction.

Bx:  $P(0)$  is " $3 \mid 4^0 - 1$ ", i.e., " $3 \mid 0$ "

which is true.

IS: Assume  $P(n)$ , i.e.,  $3|4^n - 1$ .

(wtp  $3|4^{n+1} - 1$ ) Then  $\exists m \in \mathbb{Z}$  s.t.

$$4^n - 1 = 3m. \text{ Then } 4^n = 3m + 1.$$

$$\text{Then } 4^{n+1} - 1 = 4^n \cdot 4 - 1 = (3m + 1) \cdot 4 - 1$$

$$= 12m + 4 - 1 = 12m + 3 = 3(4m + 1).$$

thus

$$3/4^{n+1} - 1$$





















